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AQA · LEVEL 2 · MATHEMATICS

AQA Functional Skills Mathematics Level 2

November 2017 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) *Pass Mark Guide: ~60%*

Specification / Paper Code: AQA-4368-QP-NOV17

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Level 2
Data Book (Examination)

Insert

Instructions

- This copy of the Data Book is for use in the examination. It should not be given to students in advance.

Advice

- This book will not be collected in for marking. Ensure that all working that you wish to have marked is written in the space provided in the question/answer book.

4368

QAN 500/8702/2

IB/M/Nov17/E5

Data Sheet for Pets Corner

At Pets Corner you can see alpacas, pigs, goats, sheep, rabbits and birds.

Buses to Pets Corner

The nearest bus stop to Pets Corner is Corner House.

Here is part of the timetable for the number 38 bus.

Whickham View	0922	0942	then at these minutes past each hour	02	22	42
General Hospital	0934	0954		14	34	54
Central Station	0941	1001		21	41	01
John Dobson Street	0952	1012		32	52	12
Sandyford Road	0955	1015		35	55	15
Corner House	1001	1021		41	01	21
Freeman Hospital	1006	1026		46	06	26

Keeping alpacas



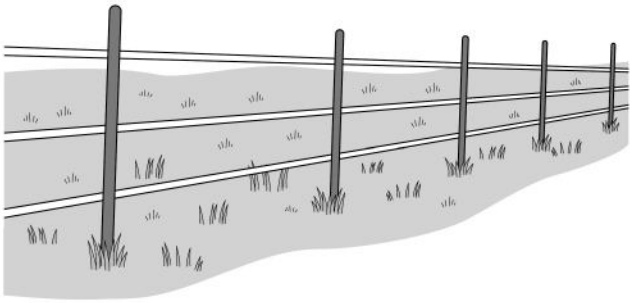
Two or three alpacas should be kept together in an enclosure.

The enclosure should have an area of at least half an acre.

Making an enclosure

Electric tape and plastic posts can be used to make a fence for an enclosure.

Here is a fence with 3 rows of electric tape.



Plastic posts are placed along each edge of the enclosure from corner to corner.

Electric tape is attached to the posts in rows.

There are usually 3 or 4 rows.

Converting an area in square metres to square yards

Use this formula.

$$y = 1.196m$$

y is the area in square yards

m is the area in square metres

Data Sheet for Water

The water that you use is measured in cubic metres or litres.
1 cubic metre = 1000 litres

With 1 cubic metre of water you can do **all** of the following.

	Water used
Have a bath	77 litres
Use a washing machine 3 times	213 litres
Have 6 showers	216 litres
Flush the toilet 32 times	288 litres
Make 49 cups of tea	12 litres
Use a dishwasher 3 times	45 litres
Cooking and other uses	149 litres
Total	**1000 litres**

Ways to save water

Have a shower instead of a bath

Put a brick in the toilet cistern

Only use the dishwasher when it is full

Ways to pay water charges for a year

With a water meter
A fixed charge
plus
an amount for each cubic metre of water used

Without a water meter
A yearly charge

Having a water meter may work out cheaper.
This depends on how much water you use.
It is free to have a water meter fitted.

END OF DATA

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Do not write outside the box

Answer **all** questions in the spaces provided.

1 **Pets Corner**

There is a **data sheet** for Pets Corner.

1 (a)



Q1 Worked Solution:
Tom
53.23
[Method & Accuracy]

Amy

Amy catches a train at 1.35 pm from Durham Station.
The train takes 16 minutes to travel to Central Station.
Amy then catches the next number 38 bus.

What time should this bus arrive at Corner House?

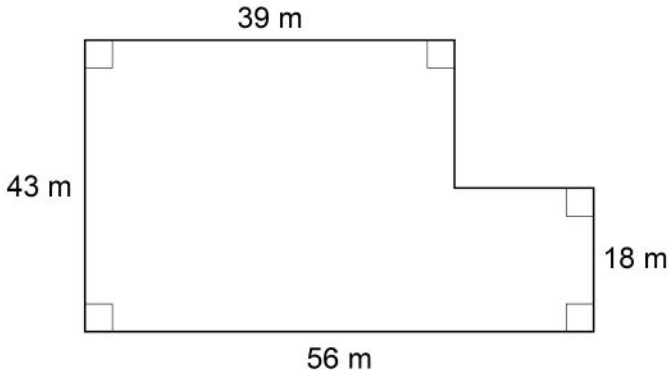
[2 marks]

Question 1 continues on the next page

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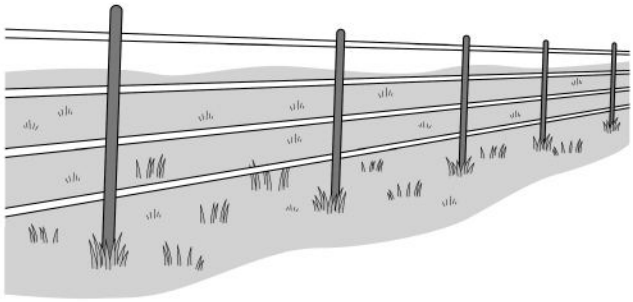
Joe is making an enclosure for alpacas.

Here is a sketch of the fence for the enclosure.



Not drawn accurately

1 (b) The fence has 4 rows of electric tape.



Electric tape is sold in rolls.
Each roll has 150 m of tape.

Joe says,

"I need to buy five rolls."

Is he correct?
You **must** show your working.

[5 marks]

Do not write outside the box

1 (c) 1 acre = 4840 square yards

Joe says,

"The enclosure will be big enough to keep three alpacas."

Is he correct?

You **must** show your working.

[6 marks]

13

Turn over for the next question

Do not write outside the box

2 Supermarket

Raj is the manager of a supermarket.

2 (a) Workers in the supermarket work in shifts.

Each day there are shifts in the morning, the afternoon and the evening.
Each person can work a maximum of two shifts each day.

Raj needs Amy, Ben, Cathy, David and Eva to work the following shifts next week.

Monday	1 worker for the evening shift
Tuesday	1 worker for the evening shift
Wednesday	1 worker for the evening shift
Thursday	2 workers for each of the three shifts
Friday	2 workers for each of the three shifts
Saturday	2 workers for each of the three shifts
Sunday	1 worker for the afternoon shift

This table shows when Amy, Ben, Cathy, David and Eva can work next week.

Q2 Worked Solution:
Zain
53.14

Amy	can only work evening shifts
Ben	can only work on Thursday, Friday and Saturday
Cathy	**cannot** work on Monday or Friday
David	can only work on Friday and Saturday
Eva	can only work on Thursday and Friday

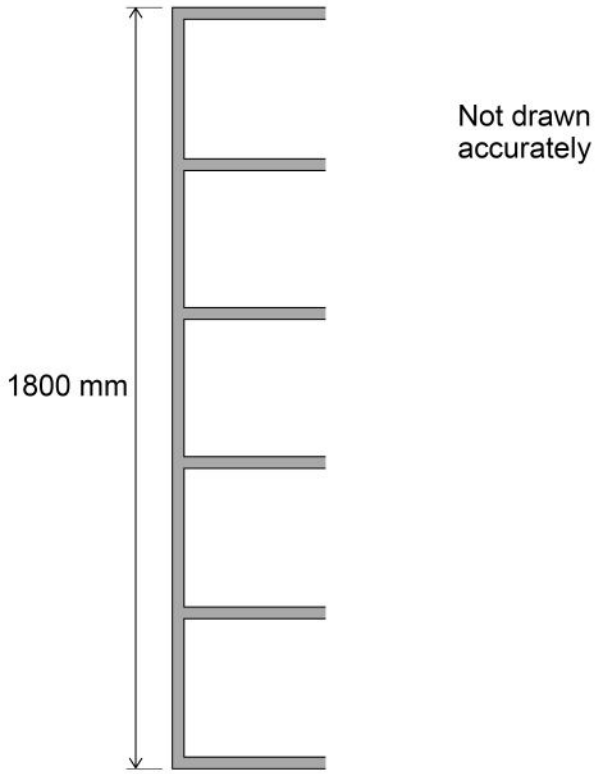
Work out a possible rota for Amy, Ben, Cathy, David and Eva for next week.
Show your rota on the opposite page.

[Method & Accuracy]

[3 marks]

Do not write outside the box

2 (b) Here is a set of shelves in the supermarket.



Each of the **six** shelves is 20 mm thick.
Each gap between the shelves is the same.

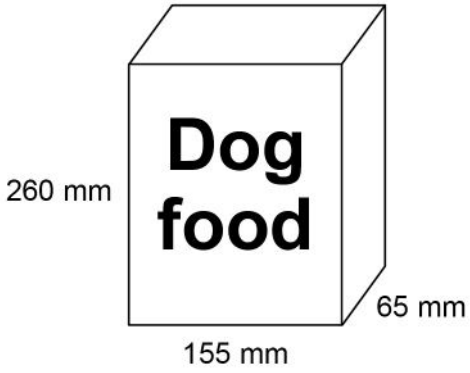
A bottle of lemonade is 320 mm high.

Will the bottle fit on the bottom shelf?
You **must** show your working.

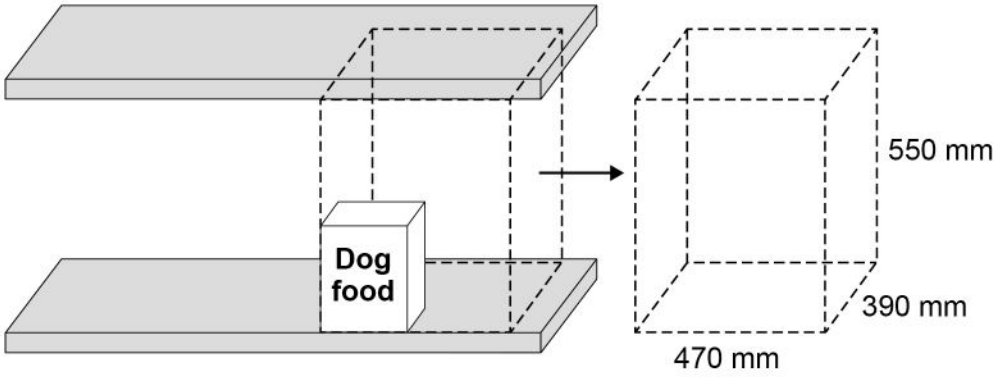
[4 marks]

Do not write outside the box

2 (c) Raj puts these boxes of dog food on another set of shelves.



He stacks the boxes in the space within the dotted lines shown on this diagram.



The space is shaped like a cuboid with

height 550 mm
width 470 mm
depth 390 mm

Work out the maximum number of boxes of dog food that can fit in the space.
You **must** show your working.

[4 marks]

Do not write outside the box

2 (d) Raj buys and sells plastic carrier bags.

He pays £11.41 per 1000 bags.
He uses this to work out the cost per bag.
He sells the bags for 5p each.

He gives 80% of the **profit** on each bag to a local charity.
One month he sold 384 carrier bags.

How much did Raj give to the charity for that month?

[5 marks]

16

Question 10 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Q10 Worked Solution:

Misread of 1.196 (e.g. 1.96) can score M4A1ft max

Not using 1.196 or misread of 1.196 can score M1M1M0M1A0 max

Examples

Area = 1983

1983 ÷ 1.196 = 1658

2420

Compares 1658 and 2420 and states or implies No

Scores A1ft as M3 scored and 1.196 used

A1ft

Area = 2408

[M2] [M0] [M1] [M1] [M0] [M1]

Do not write outside the box

3 **Water**

There is a **data sheet** for Water.

3 (a) Mia uses her dishwasher 15 times.

Circle the amount of water this uses.

[1 mark]

45 litres 57 litres 225 litres 675 litres

Q3 Worked Solution:

Paul

3 (b) Liam normally uses 1200 litres of water each week.

One week he decides to save water.

52.93

Normally, he has 3 baths each week.

He decides to have 3 showers instead.

He puts a brick in the toilet cistern.

This saves 1.5 litres of water each flush.

He flushes the toilet 30 times that week.

He says,

"This saved $\frac{1}{6}$ of the total amount of water I normally use each week."

[Method & Accuracy]

Is he correct?

You **must** show your working.

[7 marks]

Question 11 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Q11 Worked Solution:

rota clearly communicated for all
shifts for all days and all names
1a
SC1 Workers 'correctly' allocated with
shifts not specified
B1B1B1 responses
Example 1 (tabulated list)
Monday
Evening
Amy
[B1]

Do not write outside the box

Ella, Seema and Jane are going to share a house for the whole of 2018



Ella



Seema



Jane

The house does **not** have a water meter.
Altogether they will use 450 **litres** of water per day.

†3 (c) How many **cubic metres** of water will the three girls use in the whole of 2018?

[2 marks]

Check your answer.
Show how you have done your check.

[1 mark]

Question 12 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Q12 Worked Solution:

Cathy
Morning
Afternoon
Morning
Afternoon
David
Afternoon
Evening
Morning
Afternoon
[Method & Accuracy]

Do not write outside the box

3 (d) Here are the ways they can pay their water charges for the **year**.

With a water meter	**Without a water meter**
£134 plus £2.96 per cubic metre of water used	12 fixed monthly payments of £53.50

The girls want the cheapest way to pay for the water they will use.

Should they have a water meter fitted?
You **must** show your working.

[5 marks]

16

Question 13 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Q13 Worked Solution:

their 1680 ÷ 5
their 356 – 20
their 1680 cannot be 1800
336 and Yes
1b
1b
336
A1ft correct decision from their value
must score M2 or
M1M0 or M0M1 with 5 shelves instead
[M1] [Aa] [A2] [A1]

Do not write outside the box

4 Swimming

4 (a) One event in a swimming competition is the men's 100 metres freestyle.

Here are the times from the two semi-finals.

Semi-Final 1	
Name	**Time (seconds)**
Ralf	53.37
Simon	53.49
Harry	53.70
Cheng	52.97
Ahmed	52.92
Ian	53.26
Mike	53.28
Jack	52.83

Semi-Final 2	
Name	**Time (seconds)**
Tom	53.23
Zain	53.14
Paul	52.93
Kev	52.88
Dai	53.20
Greg	53.66
Yan	53.28
Louis	53.57

The eight fastest swimmers from the semi-finals go into the final.

Each swimmer in the final is given a lane number.

The table below shows how the lane numbers are decided.

Time in semi-finals	Lane number
1st fastest	4
2nd fastest	5
3rd fastest	3
4th fastest	6
5th fastest	2
6th fastest	7
7th fastest	1
8th fastest	8

Question 14 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Q14 Worked Solution:

1680 and 1600 and Yes
5.25 and Yes
Ib
Ib
1680 and 1600
5.25
A1ft correct decision from their value
must score M2 or
M1M0 or M0M1 with 5 shelves instead
of 6 or 6 spaces instead of 5
[A2] [A1] [M1] [M0] [M0] [M1]

Question 4 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Q4 Worked Solution:

Kev

52.88

[Method & Accuracy]

Do not write outside the box

Complete the table for the final.

[4 marks]

Lane number	Name
1	
2	
3	
4	
5	
6	
7	
8	

†4 (b) The winning time in the final is 52.28 seconds.
The competition record is 51.91 seconds.

How much slower is the winning time than the record time?

[1 mark]

Check your answer.
Show how you have done your check.

[1 mark]

Question 15 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

QIS Worked Solution:

their 2 × their 3 × their 6

multiplying their values

allow if incorrectly rounded or not rounded

36

These attempts are common and score zero

□ (470 × 390 × 550) ÷ (155 × 65 × 260) or 38

□ (470 + 390 + 550) ÷ (155 + 65 + 260) or 3

[M1] [Rc] [A1] [Aa]

Do not write outside the box

4 (c) Beth and Daisy are members of a swimming club.
They both swim in 100 metres backstroke races.

Here are their times, in seconds, for their races in 2017

Beth

65.7 66.6 67.4 63.6 65.2 64.8 66.5 64.9 68.5 66.8

Daisy

62.5 63.2 67.4 62.6 64.2 66.2 64.8 65.9

For the next race the club wants to choose the better swimmer.

Use the data to decide which swimmer the club should choose.
You **must** show your working.

[4 marks]

Question 16 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Q16 Worked Solution:

See official AQA mark scheme method.

[Method & Accuracy]

Do not write outside the box

4 (d) Erin is the manager of a junior swimming team.



I need to buy swimming kit for six swimmers.
I also need to hire a minibus.

Erin

She sees these adverts

Swimming kit	
Swimsuit	£46.95
Swim cap	£6.50
Tracksuit	£62.95
15% off orders over £500	

Minibus Hire
£370

Erin wants to buy each of the six swimmers a swimsuit, swim cap and tracksuit.
Show that it will cost **less than** £1000 for the kit and minibus.

[5 marks]

15

END OF QUESTIONS

Question 17 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Q17 Worked Solution:

See official AQA mark scheme method.

[Method & Accuracy]

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ANSWER IN THE SPACES PROVIDED

Question 18 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Q18 Worked Solution:

See official AQA mark scheme method.

[Method & Accuracy]

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Question 20 · Continued

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Q20 Worked Solution:

See official AQA mark scheme method.

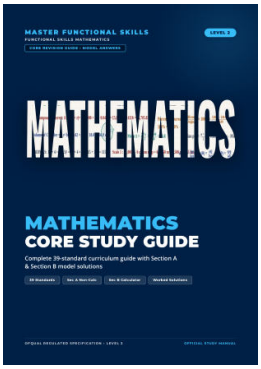
[Method & Accuracy]

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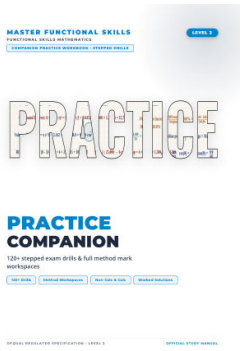


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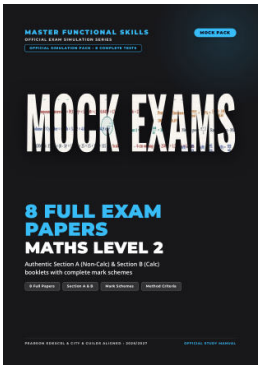


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